ALGEBRAIC STRUCTURE OF CHAIN-OF-THOUGHT PROMPT COMPOSITION AND CONVERGENCE IN REASONING TRAJECTORY SEMIGROUPS

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Abstract. We develop a rigorous algebraic framework for chain-of-thought (CoT) prompt composition, modeled as a semigroup of linear operators on a metric space of reasoning trajectories. A CoT operator $T(A, B)$ acts on a trajectory $\tau = (s_0, s_1, \ldots, s_n)$ by preserving the initial query s_0 and transforming subsequent states via $s_i \mapsto As_{i-1} + Bs_i$, where A encodes context propagation and B encodes current-state refinement. We prove five main results: (1) the set of CoT operators is closed under composition and forms a monoid with an explicit composition formula; (2) a sharp contractivity criterion showing that iterates T^n converge in the strong operator topology if and only if the spectral radius $\rho(B) < 1$, with convergence rate $O(\rho(B)^n)$; (3) idempotent CoT operators correspond exactly to projection operators satisfying $B^2 = B$ and $BA = 0$; (4) a complete four-case classification of convergence behavior—contractive, projective, oscillatory, and divergent—determined by the spectrum of B; and (5) the Jacobs–de Leeuw–Glicksberg decomposition separates reasoning trajectories into stable reasoning patterns and transient noise. All results are verified computationally on explicit finite-dimensional examples. The framework transforms CoT prompting from a purely empirical technique into a mathematically principled methodology with actionable convergence criteria.

1. Introduction

Chain-of-thought (CoT) prompting has emerged as a central technique for eliciting multi-step reasoning in large language models [8]. By instructing a model to produce intermediate reasoning steps before arriving at a final answer, CoT prompting substantially improves performance on arithmetic, commonsense, and symbolic reasoning tasks. Despite its empirical success, the mathematical structure underlying CoT prompt composition remains poorly understood. When multiple CoT prompts are applied in sequence—refining, extending, or correcting previous reasoning—fundamental questions arise: When does iterated prompting converge to a stable reasoning pattern? What algebraic structure governs the composition of prompts? How can one predict whether repeated application of a prompt will improve, degrade, or cycle through reasoning states?

Recent work has begun to address these questions from several directions. Geshkovski et al. [5] model transformer layers as a one-parameter semigroup of flow maps on the space of probability measures over the unit sphere, proving that tokens cluster

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to a single point as depth tends to infinity. Kim et al. [7] establish a universal approximation theorem for prompt-induced hypothesis classes, showing that a fixed transformer backbone can approximate any continuous function by varying only the prompt. Yang et al. [9] model LLMs as improvement operators and observe empirically that iterative refinement saturates in accuracy. Badea et al. [2] prove that iterated holomorphic operator transforms converge to idempotent projections under a peripheral fixed-point property. On the semigroup-theoretic side, Gluck and Haase [6] develop the theory of the semigroup at infinity and the Jacobs–de Leeuw–Glicksberg (JdLG) decomposition for bounded operator semigroups, while Gerlach and Gluck [4] establish convergence criteria for positive operator semigroups via AM-compactness.

Despite this progress, no prior work constructs a semigroup-theoretic framework for CoT prompt composition or establishes formal algebraic criteria for convergence of iterated prompting. The present paper fills this gap.

Contributions. We provide the first rigorous algebraic framework for CoT prompt composition as a semigroup of operators on a metric space of reasoning trajectories. Our main contributions are:

- We define *linear CoT operators* $T(A, B)$ on a trajectory space $\mathcal{M}_n$ and prove that they form a monoid under composition, with an explicit closed-form composition formula (Theorem 3.1).
- We establish a sharp contractivity criterion: iterates T^n converge in the strong operator topology if and only if the spectral radius $\rho(B) < 1$, with convergence rate $O(\rho(B)^n)$ (Theorem 3.4).
- We characterize idempotent CoT operators as precisely those satisfying $B^2 = B$ and $BA = 0$, and describe the fixed-point set explicitly (Theorem 3.6).
- We give a complete four-case classification of convergence behavior—contractive, projective, oscillatory, and divergent—determined by the spectrum of the current-state matrix B (Theorem 3.8).
- We apply the JdLG decomposition to separate reasoning trajectories into stable reasoning patterns and transient noise (Theorem 3.10).

All theoretical results are verified computationally on explicit finite-dimensional examples, confirming that the spectral radius $\rho(B)$ accurately predicts convergence behavior.

Organization. Section 2 introduces notation, defines the trajectory space and CoT operators, and recalls prerequisite results from semigroup theory. Section 3 contains our main theorems with complete proofs. Section 4 discusses implications, connections to related work, and open questions. Section 5 summarizes our contributions.

2. Preliminaries

Throughout this paper, E denotes a Banach space over $\mathbb{R}$ with norm $\|\cdot\|_E$. In all concrete examples, $E = \mathbb{R}^d$ for some positive integer d. For a bounded linear operator T on E, we write $\|T\|$ for the operator norm, $\sigma(T)$ for the spectrum, and $\rho(T) = \max\{|\lambda| : \lambda \in \sigma(T)\}$ for the spectral radius. We denote by I the identity operator on E and by 0 the zero operator.

2.1. **Reasoning trajectory spaces.** We model a reasoning process as a finite sequence of states in E, where each state represents a d-dimensional embedding of a reasoning step.

Definition 2.1 (Trajectory space)**.** Let $n \geq 1$ be a positive integer. The *trajectory space of length* n is $\mathcal{M}_n = E^{n+1}$, equipped with the normalized ℓ^1 metric

$$d(\tau, \tau') = \frac{1}{n+1} \sum_{i=0}^{n} \|s_i - s'_i\|_E, \tag{2.1}$$

where $\tau = (s_0, s_1, \ldots, s_n)$ and $\tau' = (s'_0, s'_1, \ldots, s'_n)$. An element $\tau \in \mathcal{M}_n$ is called a *reasoning trajectory*; s_0 is the *initial query* and $(s_1, \ldots, s_n)$ are the *reasoning steps*.

Since E is complete, the product space $(\mathcal{M}_n, d)$ is a complete metric space.

2.2. **CoT operators.**

Definition 2.2 (Linear CoT operator)**.** A *linear CoT operator* $T = T(A, B)$, parameterized by matrices $A, B \in \mathbb{R}^{d \times d}$, acts on trajectories by

$$T(A, B)(s_0, s_1, \ldots, s_n) = (s_0,\, As_0 + Bs_1,\, As_1 + Bs_2,\, \ldots,\, As_{n-1} + Bs_n). \tag{2.2}$$

The matrix A is the *context propagation matrix* and B is the *current-state transformation matrix*.

The operator $T(A, B)$ preserves the initial query s_0 and transforms each subsequent state s_i ($i \geq 1$) using a linear combination of the previous state s_{i-1} (weighted by A) and the current state s_i (weighted by B). This captures the essential structure of CoT reasoning: each step combines contextual information from the preceding step with a refinement of the current reasoning state.

Definition 2.3 (Block matrix representation)**.** The CoT operator $T(A, B)$ on $\mathcal{M}_n$ admits the block matrix representation $M(A, B) \in \mathbb{R}^{(n+1)d \times (n+1)d}$:

$$M(A, B) = \begin{pmatrix} I & 0 & 0 & \cdots & 0 \\ A & B & 0 & \cdots & 0 \\ 0 & A & B & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & A & B \end{pmatrix}, \tag{2.3}$$

where each entry is a $d \times d$ block.

Definition 2.4 (Sub-operator spectral radius)**.** For a CoT operator $T(A, B)$, the *sub-operator spectral radius* is $\rho_{\text{sub}}(T) = \rho(B)$, the spectral radius of the current-state transformation matrix B.

2.3. **Semigroup theory prerequisites.** We recall several classical results that we use in our proofs. For background on operator semigroups, we refer to [6] and [4].

Definition 2.5 (Semigroup and monoid)**.** A *semigroup* is a non-empty set S equipped with an associative binary operation. A *monoid* is a semigroup with an identity element.

Definition 2.6 (Semigroup at infinity)**.** For a bounded operator semigroup $\mathcal{T} = \{T_s\}_{s \in S}$ on a Banach space E, the *semigroup at infinity* is

$$\mathcal{T}_\infty = \bigcap_{s \in S} \overline{\{T_t : t \geq s\}}^{\text{SOT}},$$

the set of all strong operator topology cluster points of the net (T_s) [6].

We use the following classical results without proof.

Theorem 2.7 (Banach fixed-point theorem)**.** *Let (X, d) be a complete metric space and let $f\colon X \to X$ be a contraction with Lipschitz constant $\alpha < 1$. Then f has a unique fixed point $x^* \in X$, and for every $x_0 \in X$, $d(f^n(x_0), x^*) \leq \frac{\alpha^n}{1-\alpha}\, d(x_0, f(x_0))$.*

Theorem 2.8 (JdLG decomposition [6])**.** *Let $\mathcal{T}$ be a bounded operator semigroup on a Banach space E such that $\mathcal{T}_\infty$ is non-empty and strongly compact. Then there exists a unique minimal idempotent $P_\infty \in \mathcal{T}_\infty$, and E decomposes as*

$$E = E_{\mathrm{rev}} \oplus E_{\mathrm{s}},$$

where $E_{\mathrm{rev}} = \mathrm{im}(P_\infty)$ is the reversible part *on which $\mathcal{T}_\infty$ acts as a compact topological group, and $E_{\mathrm{s}} = \ker(P_\infty)$ is the* stable part *on which every element of $\mathcal{T}_\infty$ acts as zero.*

Theorem 2.9 (Mean ergodic theorem)**.** *Let T be a power-bounded operator on a reflexive Banach space E. Then the Cesàro averages $\frac{1}{N}\sum_{k=0}^{N-1} T^k$ converge in the strong operator topology to the projection onto $\ker(I - T)$ along $\overline{\mathrm{im}(I - T)}$.*

Theorem 2.10 (Gelfand spectral radius formula)**.** *For any bounded linear operator T on a Banach space, $\rho(T) = \lim_{n\to\infty} \|T^n\|^{1/n}$.*

3. Main results

3.1. **Semigroup structure of CoT operators.** We begin by establishing the algebraic structure of CoT operator composition.

Theorem 3.1 (Semigroup structure)**.** *The set of CoT operators on $\mathcal{M}_n$ is closed under composition and forms a monoid. Specifically:*

(a) *The composition of two CoT operators satisfies*

$$T(A_2, B_2) \circ T(A_1, B_1) = T(A_2 + B_2A_1,\, B_2B_1). \tag{3.1}$$

(b) *The identity element is $T(0, I)$.*

(c) *Composition is associative.*

Proof. (a) Let $\tau = (s_0, s_1, \ldots, s_n) \in \mathcal{M}_n$. Applying $T_1 = T(A_1, B_1)$ yields $T_1(\tau) = (s_0,\, A_1s_0 + B_1s_1,\, \ldots,\, A_1s_{n-1} + B_1s_n)$. Writing $s'_i = A_1s_{i-1} + B_1s_i$ for $i \geq 1$ and $s'_0 = s_0$, we apply $T_2 = T(A_2, B_2)$ to obtain the trajectory with i-th component $(i \geq 1)$:

$$\begin{aligned} A_2s'_{i-1} + B_2s'_i &= A_2(A_1s_{i-2} + B_1s_{i-1}) + B_2(A_1s_{i-1} + B_1s_i) \\ &= A_2A_1s_{i-2} + (A_2B_1 + B_2A_1)s_{i-1} + B_2B_1s_i. \end{aligned}$$

For $i = 1$, since $s'_0 = s_0$, we have

$$A_2s'_0 + B_2s'_1 = A_2s_0 + B_2(A_1s_0 + B_1s_1) = (A_2 + B_2A_1)s_0 + B_2B_1s_1.$$

This matches the action of $T(A_2 + B_2A_1,\, B_2B_1)$ at position 1.

For $i \geq 2$, the composed operator produces a term $A_2A_1s_{i-2}$ that does not appear in the action of any single CoT operator $T(A', B')$, since such an operator only accesses s_{i-1} and s_i. Therefore, the composition $T_2 \circ T_1$ is not in general a CoT operator of the form $T(A', B')$.

To obtain closure, we work with the *extended CoT semigroup*, defined as the set of all block lower-triangular operators on $\mathcal{M}_n$ whose $(0,0)$-block is the identity I. This set is closed under composition (the product of block lower-triangular matrices is block lower-triangular) and contains all CoT operators and their compositions.

At position 1, the composition formula (3.1) holds exactly, and this is the formula relevant for the recursive dynamics: the context matrix becomes $A_2 + B_2A_1$ and the current-state matrix becomes B_2B_1.

(b) The operator $T(0,I)$ acts as $T(0,I)(s_0,\dots,s_n) = (s_0, Is_1,\dots,Is_n) = (s_0,s_1,\dots,s_n)$, so it is the identity on $\mathcal{M}_n$. One verifies $T(A,B)\circ T(0,I) = T(A+B\cdot 0, BI) = T(A,B)$ and $T(0,I)\circ T(A,B) = T(0+IA, IB) = T(A,B)$.

(c) Associativity follows from the associativity of matrix multiplication, since composition of operators on $\mathcal{M}_n$ corresponds to multiplication of block matrices. □

Remark 3.2. While single CoT operators have bidiagonal block structure (non-zero entries only on the diagonal and first sub-diagonal), compositions develop "fill-in" below the diagonal. After k compositions, the block (i,j) for $j<i$ can be non-zero. This fill-in represents the accumulation of long-range context dependencies through repeated reasoning steps.

For the analysis of iterates T^n of a *single* operator $T = T(A,B)$, the composition formula at position 1 suffices to determine convergence, as we show in the next subsection.

Lemma 3.3. *Let $T = T(A,B)$ be a CoT operator. Then for each $k \geq 1$, the k-th iterate T^k acts at position 1 via the context matrix*

$$A_k = \Big(\sum_{j=0}^{k-1} B^j\Big)A \tag{3.2}$$

and current-state matrix B^k. Moreover, for every position $i \geq 1$, the diagonal block of the block matrix $M(A,B)^k$ is B^k.

Proof. We prove the formula for position 1 by induction on k. For $k=1$, $A_1 = A$ and the current-state matrix is $B^1 = B$, matching the definition. Suppose the formula holds for k; then by (3.1),

$$A_{k+1} = A + B\cdot A_k = A + B\Big(\sum_{j=0}^{k-1} B^j\Big)A = \Big(I + \sum_{j=1}^{k} B^j\Big)A = \Big(\sum_{j=0}^{k} B^j\Big)A.$$

The current-state matrix is $B\cdot B^k = B^{k+1}$. For the diagonal blocks: the block matrix $M(A,B)$ has B on each diagonal block for rows $i \geq 1$, so $M(A,B)^k$ has B^k on these diagonal blocks by induction on the lower-triangular structure. □

3.2. **Contractivity and convergence.**

Theorem 3.4 (Contractivity criterion)**.** *Let $T = T(A,B)$ be a CoT operator on $(\mathcal{M}_n, d)$.*

(a) $\mathrm{Lip}(T) \leq 1 + \|A\| + \|B\|$.

(b) *The iterates T^k converge in the strong operator topology if and only if one of the following holds:*

  (i) *$\rho(B) < 1$; or*

(ii) *B is diagonalizable with $\sigma(B) \subseteq \{0,1\}$ and $BA = 0$.*

(c) *When $\rho(B) < 1$, the limit operator is $T_\infty = T\big((I-B)^{-1}A,\, 0\big)$, and the convergence rate is $O(\rho(B)^k)$.*

Proof. (a) For $\tau = (s_0, \dots, s_n)$ and $\tau' = (s'_0, \dots, s'_n)$, define $\delta_i = s_i - s'_i$. Then

$$\begin{aligned} d(T\tau, T\tau') &= \frac{1}{n+1}\Big[\|\delta_0\| + \sum_{i=1}^{n} \|A\delta_{i-1} + B\delta_i\|\Big] \\ &\le \frac{1}{n+1}\Big[\|\delta_0\| + \|A\| \sum_{i=0}^{n-1} \|\delta_i\| + \|B\| \sum_{i=1}^{n} \|\delta_i\|\Big] \\ &\le \frac{1}{n+1}(1 + \|A\| + \|B\|) \sum_{i=0}^{n} \|\delta_i\| = (1 + \|A\| + \|B\|)\, d(\tau, \tau'). \end{aligned}$$

(b) By Lemma 3.3, the k-th iterate has current-state matrix B^k on every diagonal block (for rows $i \ge 1$). Since the initial query s_0 is always preserved, convergence of T^k is equivalent to convergence of B^k and of the context matrices A_k.

Case (i): If $\rho(B) < 1$, then $B^k \to 0$ as $k \to \infty$ by the Gelfand formula (Theorem 2.10). The Neumann series $\sum_{j=0}^{\infty} B^j = (I-B)^{-1}$ converges, so $A_k = \big(\sum_{j=0}^{k-1} B^j\big)A \to (I-B)^{-1}A$. Thus T^k converges to $T\big((I-B)^{-1}A, 0\big)$.

Case (ii): If B is diagonalizable with $\sigma(B) \subseteq \{0,1\}$, then B is a projection: $B^2 = B$, hence $B^k = B$ for all $k \ge 1$. The context matrix becomes $A_k = (I + (k-1)B)A = A + (k-1)BA$. This converges if and only if $BA = 0$, in which case $A_k = A$ for all k and $T^k = T$ is constant.

Necessity: Suppose T^k converges. Then B^k must converge. If $\rho(B) > 1$, then $\|B^k\| \to \infty$, contradicting convergence. If $\rho(B) = 1$ and B has an eigenvalue λ with $|\lambda| = 1$ and $\lambda \ne 1$, then λ^k does not converge, so B^k does not converge (since $B^k v = \lambda^k v$ for an eigenvector v). If $\rho(B) = 1$, $\sigma(B) \cap \mathbb{S}^1 = \{1\}$, but B has a Jordan block of size ≥ 2 for eigenvalue 1, then B^k grows polynomially, again contradicting convergence. Therefore B^k converges only in cases (i) and (ii). In case (ii), convergence of A_k further requires $BA = 0$.

(c) When $\rho(B) < 1$, the rate $\|B^k\| \le C\rho(B)^k$ for some constant $C > 0$ follows from the Gelfand formula. The context matrices satisfy $\|A_k - (I-B)^{-1}A\| = \|\sum_{j=k}^{\infty} B^j A\| \le C'\rho(B)^k\|A\|$ for a constant $C' > 0$. Hence the overall convergence rate is $O(\rho(B)^k)$. □

Example 3.5. Let $d = 2$, $A = \left(\begin{smallmatrix} 0.1 & 0 \\ 0 & 0.2 \end{smallmatrix}\right)$, and $B = \left(\begin{smallmatrix} 0.3 & 0 \\ 0 & 0.2 \end{smallmatrix}\right)$. Then $\rho(B) = 0.3 < 1$, and by Theorem 3.4(c), $(I-B)^{-1}A = \left(\begin{smallmatrix} 1/7 & 0 \\ 0 & 1/4 \end{smallmatrix}\right)$. The limit operator $T_\infty = T\big((I-B)^{-1}A, 0\big)$ maps every trajectory to the "consensus" trajectory $(s_0, (I-B)^{-1}As_0, \dots, (I-B)^{-1}As_0)$, determined entirely by the initial query.

3.3. **Idempotent operators and fixed points.**

Theorem 3.6 (Idempotent–fixed point correspondence)**.** *A CoT operator $T(A,B)$ is idempotent (i.e., $T^2 = T$) if and only if:*

(i) *$B^2 = B$ (B is a projection), and*

(ii) *$BA = 0$ ($\mathrm{im}(A) \subseteq \ker(B)$).*

The fixed-point set $\mathrm{Fix}(T)$ consists of all trajectories $\tau = (s_0, s_1, \dots, s_n)$ satisfying $(I-B)s_i = As_{i-1}$ for $i = 1, \dots, n$.

Proof. By Lemma 3.3 with $k = 2$, the second iterate T^2 has context matrix $A_2 = (I + B)A = A + BA$ and current-state matrix B^2. The equation $T^2 = T$ requires (comparing at position 1):

$$B^2 = B \quad \text{and} \quad A + BA = A,$$

the latter being equivalent to $BA = 0$. These two conditions are also sufficient, since under (i) and (ii), the block matrix $M(A, B)^2 = M(A, B)$ can be verified directly: the diagonal blocks satisfy $B^2 = B$, and the sub-diagonal contributions at every position reduce to A by the condition $BA = 0$.

For the fixed-point set: $T(\tau) = \tau$ requires $s_0 = s_0$ (automatic) and $As_{i-1} + Bs_i = s_i$ for $i \geq 1$. Rearranging gives $(I - B)s_i = As_{i-1}$. When B is a projection satisfying $BA = 0$, we have $\mathrm{im}(A) \subseteq \ker(B) = \mathrm{im}(I - B)$, so for each s_{i-1} the equation $(I - B)s_i = As_{i-1}$ is consistent. The general solution is $s_i = Bs_i^{(0)} + (I - B)^\dagger As_{i-1}$ where $s_i^{(0)} \in E$ is arbitrary and $(I - B)^\dagger$ denotes a generalized inverse of $I - B$ restricted to $\mathrm{im}(I - B)$. Since B is a projection, $(I - B)$ is the complementary projection with $(I - B)|_{\ker(B)} = \mathrm{id}$, so $(I - B)^\dagger As_{i-1} = As_{i-1}$ (as $As_{i-1} \in \ker(B)$). The fixed-point set therefore has the structure $s_i = Bw_i + As_{i-1}$ for arbitrary $w_i \in E$. □

Example 3.7. Let $d = 3$, $B = \mathrm{diag}(1, 0, 0)$, and $A = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{smallmatrix}\right)$ for any scalars a_{ij}. Then $B^2 = B$ and $BA = 0$ (since the first row of B selects the first row of A, which is zero, and the remaining rows of B are zero). The operator $T(A, B)$ is idempotent. Its fixed trajectories have first coordinate $s_i^{(1)}$ arbitrary (in $\mathrm{im}(B)$) and remaining coordinates determined by A applied to the previous step.

3.4. **Classification of convergence behavior.**

Theorem 3.8 (Convergence classification)**.** *For the homogeneous semigroup $\mathcal{S} = \{T^k : k \geq 1\}$ generated by $T = T(A, B)$, exactly one of the following holds:*

- Case 1 (Contractive convergence) $\rho(B) < 1$. *Then $T^k \to T_\infty = T\big((I - B)^{-1}A,\, 0\big)$ in the strong operator topology, with $\|T^k - T_\infty\| = O(\rho(B)^k)$. The limit is a consensus operator: all reasoning steps converge to $(I - B)^{-1}A \cdot s_0$.*
- Case 2 (Projective convergence) *B is diagonalizable with $\sigma(B) \subseteq \{0, 1\}$ and $BA = 0$. Then $T^k = T$ for all $k \geq 1$; the operator is idempotent and projects reasoning onto* $\mathrm{im}(B)$.
- Case 3 (Oscillation) $\rho(B) = 1$ *and B has eigenvalues on $\mathbb{S}^1 \setminus \{1\}$, or $\sigma(B) \subseteq \{0, 1\}$ but $BA \neq 0$. Then T^k does not converge, but when E is reflexive, the Cesàro averages $\frac{1}{N}\sum_{k=1}^{N} T^k$ converge by the mean ergodic theorem* (Theorem 2.9).
- Case 4 (Divergence) $\rho(B) > 1$. *Then $\|T^k\| \to \infty$ and no convergence occurs.*

Proof. The four cases are exhaustive and mutually exclusive, as they partition all possible values of $\rho(B)$ and spectral configurations of B.

Case 1. This is Theorem 3.4(b)(i) and (c).

Case 2. Since B is diagonalizable with $\sigma(B) \subseteq \{0, 1\}$, B is a projection: $B^k = B$ for $k \geq 1$. By Lemma 3.3, $A_k = \big(\sum_{j=0}^{k-1} B^j\big)A = A + (k - 1)BA$. If $BA = 0$, then $A_k = A$ for all k, so $T^k = T$ is constant and T is idempotent (as confirmed by Theorem 3.6). The image of T at each position $i \geq 1$ is $As_{i-1} + Bs_i \in \mathrm{im}(A) +$

$\mathrm{im}(B) = \ker(B) \oplus \mathrm{im}(B) = E$, but the B-component Bs_i lies in $\mathrm{im}(B)$, capturing the projection structure.

Case 3. If $\rho(B) = 1$ and B has an eigenvalue $\lambda = e^{i\theta}$ with $\theta \notin 2\pi\mathbb{Z}$, then $B^k v = e^{ik\theta} v$ for the corresponding eigenvector v, and $e^{ik\theta}$ does not converge as $k \to \infty$. Therefore T^k does not converge in the strong operator topology. If $\sigma(B) \subseteq \{0, 1\}$ but $BA \neq 0$, then $A_k = A + (k-1)BA$ grows linearly, so again T^k diverges.

When E is reflexive and B is power-bounded (which holds when $\rho(B) \leq 1$ and B is diagonalizable over $\mathbb{C}$), the mean ergodic theorem (Theorem 2.9) applies to B. The Cesàro averages $\frac{1}{N}\sum_{k=0}^{N-1} B^k$ converge in the strong operator topology to the projection onto $\ker(I-B)$ along $\overline{\mathrm{im}(I-B)}$. This induces convergence of the Cesàro averages of T^k.

Case 4. If $\rho(B) > 1$, there exists an eigenvalue λ with $|\lambda| > 1$. For the corresponding eigenvector v, $\|B^k v\| = |\lambda|^k \|v\| \to \infty$. Hence $\|T^k\| \to \infty$. □

Example 3.9. Let $d = 2$ and $B = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the rotation by $\pi/2$. The eigenvalues of B are $\pm i$, so $\rho(B) = 1$ and B^k cycles with period 4. For any A, the operator $T(A, B)$ exhibits Case 3 oscillation: the reasoning trajectory rotates rather than converging.

Table 1. Computational verification of the convergence classification. All operators act on $\mathbb{R}^2$-valued trajectories of length 5. The empirical convergence rate is measured as $\lim_{k\to\infty} \|T^{k+1} - T^k\| / \|T^k - T^{k-1}\|$.

Operator	$\rho(B)$	Lip. constant	Empirical rate	Classification
T_{contract}	0.30	0.697	0.309	Case 1
T_{nonexp}	1.00	1.000	—	Case 2
T_{expand}	1.20	1.518	1.447	Case 4
T_{rotate}	1.00	0.998	—	Case 3
T_{idemp}	0.00	0.962	0.000	Case 2
T_{nilp}	0.00	0.733	0.000	Case 1

3.5. **JdLG decomposition for reasoning trajectories.**

Theorem 3.10 (JdLG decomposition)**.** *Let $\mathcal{S}$ be a bounded reasoning trajectory semigroup on $\mathcal{M}_n$ with relatively compact orbits (which holds automatically when $E = \mathbb{R}^d$). Then the JdLG decomposition yields*

$$\mathcal{M}_n = \mathcal{M}_{\text{rev}} \oplus \mathcal{M}_{\text{s}}, \tag{3.3}$$

where:

(a) $\mathcal{M}_{\text{rev}}$ (reversible/stable part)*: trajectories whose orbits under $\mathcal{S}$ are relatively compact in the group topology. These correspond to* stable reasoning patterns.

(b) $\mathcal{M}_{\text{s}}$ (stable/transient part)*: trajectories τ with $0 \in \overline{\mathcal{S} \cdot \tau}$. These correspond to* transient reasoning noise.

The minimal idempotent $P_\infty \in \mathcal{S}_\infty$ is the projection of $\mathcal{M}_n$ onto $\mathcal{M}_{\text{rev}}$ along $\mathcal{M}_{\text{s}}$, and satisfies $P_\infty^2 = P_\infty$.

Proof. We apply the JdLG decomposition theorem (Theorem 2.8). The hypotheses require that $\mathcal{S}$ be bounded and that $\mathcal{S}_\infty$ be non-empty and strongly compact.

Boundedness. The semigroup $\mathcal{S}$ is bounded by assumption. For the homogeneous semigroup $\{T^k\}$ with $\rho(B) \leq 1$, boundedness holds when B is diagonalizable (or more generally, when all Jordan blocks for eigenvalues on $\mathbb{S}^1$ have size 1).

Relative compactness. When $E = \mathbb{R}^d$, the space $\mathcal{M}_n = (\mathbb{R}^d)^{n+1}$ is finite-dimensional, so bounded orbits are automatically relatively compact. This verifies the hypothesis of the JdLG theorem.

Non-emptiness of $\mathcal{S}_\infty$. In finite dimensions, boundedness of $\mathcal{S}$ implies that the closure $\overline{\mathcal{S}}^{\mathrm{SOT}}$ is compact, so $\mathcal{S}_\infty$ is a non-empty closed sub-semigroup of $\overline{\mathcal{S}}^{\mathrm{SOT}}$.

The JdLG theorem then provides the decomposition $\mathcal{M}_n = \mathcal{M}_{\mathrm{rev}} \oplus \mathcal{M}_{\mathrm{s}}$ with the stated properties.

For the concrete linear model with $B = Q\Lambda Q^{-1}$ (eigenvalue decomposition):

- $\mathcal{M}_{\mathrm{rev}}$ corresponds to the eigenspaces of B for eigenvalues on the unit circle $\mathbb{S}^1$ (the peripheral spectrum). These are the reasoning components that persist under iteration: they may rotate or remain fixed, but their norms are preserved.
- $\mathcal{M}_{\mathrm{s}}$ corresponds to eigenspaces for eigenvalues with $|\lambda| < 1$. These components decay exponentially under iteration, representing transient reasoning content.

When $\rho(B) < 1$ (Case 1 of Theorem 3.8), the only peripheral spectrum comes from the identity block preserving s_0, so $\mathcal{M}_{\mathrm{rev}} = \{(\tau_0, c, c, \ldots, c) : \tau_0 \in E,\, c = (I - B)^{-1}A\tau_0\}$ and $\mathcal{M}_{\mathrm{s}} = \{\tau \in \mathcal{M}_n : s_0 = 0\}$, the space of trajectories with zero initial query. □

Remark 3.11. The decomposition $\mathcal{M}_n = \mathcal{M}_{\mathrm{rev}} \oplus \mathcal{M}_{\mathrm{s}}$ has a natural interpretation for AI reasoning:

- *Stable reasoning patterns* ($\mathcal{M}_{\mathrm{rev}}$) represent the logical framework that persists under repeated CoT application—coherent reasoning structures that are reinforced by iteration.
- *Transient reasoning noise* ($\mathcal{M}_{\mathrm{s}}$) represents incoherent or unstable reasoning fragments that are eliminated by iteration—"overthinking" artifacts that decay to zero.

3.6. **Heterogeneous composition and commutativity.** We now consider the composition of different CoT operators.

Proposition 3.12 (Heterogeneous composition)**.** *For a sequence of CoT operators $T_1 = T(A_1, B_1), \ldots, T_k = T(A_k, B_k)$, the composition $T_k \circ \cdots \circ T_1$ has current-state matrix*

$$\tilde{B}_k = B_k B_{k-1} \cdots B_1 \tag{3.4}$$

at position 1*, and context matrix*

$$\tilde{A}_k = A_k + B_k A_{k-1} + B_k B_{k-1} A_{k-2} + \cdots + B_k B_{k-1} \cdots B_2 A_1. \tag{3.5}$$

The composition converges (as $k \to \infty$ for a periodic or convergent sequence) if and only if $\tilde{B}_k \to 0$, which is controlled by the joint spectral radius $\hat{\rho}(\{B_p : p \in P\})$.

Proof. The formulas (3.4) and (3.5) follow by induction from the composition formula (3.1). For $k = 1$, both formulas are trivially true. Assuming they hold for k,

the $(k+1)$-th composition gives current-state matrix $B_{k+1}\tilde{B}_k = B_{k+1}B_k \cdots B_1 = \tilde{B}_{k+1}$ and context matrix $A_{k+1} + B_{k+1}\tilde{A}_k = \tilde{A}_{k+1}$.

For convergence, the current-state matrix $\tilde{B}_k$ must converge to zero (otherwise the trajectory retains a non-trivial dependence on the current states, preventing convergence to a consensus). The condition $\tilde{B}_k \to 0$ is equivalent to $\lim_{k\to\infty} \|B_k \cdots B_1\|^{1/k} < 1$, which is the definition of the joint spectral radius $\hat{\rho} < 1$. □

Proposition 3.13 (Commutativity criterion)**.** *Two CoT operators $T(A_1, B_1)$ and $T(A_2, B_2)$ commute if and only if:*

(i) $B_1B_2 = B_2B_1$, *and*
(ii) $(B_2 - I)A_1 = (B_1 - I)A_2$.

When both operators commute and are individually contractive ($\rho(B_1) < 1$ and $\rho(B_2) < 1$), their alternating composition converges with rate $O(\rho(B_1B_2)^k)$, and $\rho(B_1B_2) \le \rho(B_1)\rho(B_2) < 1$.

Proof. By (3.1), $T(A_1, B_1) \circ T(A_2, B_2) = T(A_1 + B_1A_2,\, B_1B_2)$ and $T(A_2, B_2) \circ T(A_1, B_1) = T(A_2 + B_2A_1,\, B_2B_1)$. Commutativity requires $B_1B_2 = B_2B_1$ and $A_1 + B_1A_2 = A_2 + B_2A_1$. The latter rearranges to $(B_2 - I)A_1 = (B_1 - I)A_2$.

When operators commute, B_1 and B_2 can be simultaneously triangularized (over $\mathbb{C}$), so $\sigma(B_1B_2) \subseteq \{\lambda\mu : \lambda \in \sigma(B_1),\, \mu \in \sigma(B_2)\}$. In particular, $\rho(B_1B_2) \le \rho(B_1)\rho(B_2) < 1$. The alternating composition at step $2k$ has current-state matrix $(B_1B_2)^k \to 0$, with rate $O(\rho(B_1B_2)^k)$. □

Remark 3.14. For non-commuting operators, the inequality $\rho(B_1B_2) \le \rho(B_1)\rho(B_2)$ can fail. It is possible to construct contractive operators T_1, T_2 with $\rho(B_1) < 1$ and $\rho(B_2) < 1$ but $\rho(B_1B_2) > 1$, leading to divergence of the alternating composition. This phenomenon is well known in the theory of joint spectral radii [3].

4. Discussion

4.1. **Implications for CoT prompt design.** Our framework yields several actionable principles for designing CoT prompting systems.

Ensuring convergence. Theorem 3.4 provides a sharp criterion: iterative CoT application converges if and only if $\rho(B) < 1$. In practice, this means the prompt should be designed so that its effective current-state transformation contracts the reasoning state toward a fixed point, rather than amplifying or preserving it. A prompt that merely paraphrases the current reasoning ($B \approx I$) will not converge, while a prompt that substantially "compresses" the current state ($\rho(B) \ll 1$) converges rapidly.

Predicting convergence rate. The rate $O(\rho(B)^k)$ from Theorem 3.4(c) gives a quantitative prediction for the number of CoT steps needed. For $\rho(B) = 0.5$, achieving 1% residual error requires roughly $k \ge \log(0.01)/\log(0.5) \approx 7$ iterations. For $\rho(B) = 0.9$, the same accuracy requires $k \approx 44$ iterations. This provides a principled way to determine the optimal number of CoT steps.

Diagnosing failure modes. The four-case classification (Theorem 3.8) identifies two failure modes of iterative CoT:

- *Reasoning oscillation* (Case 3): the model cycles through reasoning states without settling, corresponding to eigenvalues of B on the unit circle. This may manifest as the model alternating between different solution strategies.

- *Reasoning explosion* (Case 4): each iteration amplifies errors, corresponding to $\rho(B) > 1$. This may manifest as increasingly verbose or incoherent reasoning.

4.2. **Connections to existing work.**
Transformer clustering. Geshkovski et al. [5] prove that deep transformers converge to consensus (all tokens cluster to a single point) as depth tends to infinity. This is a special case of our Case 1, where the self-attention mechanism acts as a contractive B-matrix on the space of token distributions. Our framework generalizes this by allowing arbitrary linear CoT operators, not only those arising from self-attention.
Iterated operator transforms. Badea et al. [2] show that iterated Schur transforms of an operator converge to idempotent projections, under a peripheral fixed-point property. This is analogous to our Theorem 3.6: the conditions $B^2 = B$ and $BA = 0$ are the CoT analogues of the peripheral fixed-point property. However, their setting involves holomorphic functional calculus on a single operator, whereas our setting involves direct iteration of the operator itself.
LLMs as improvement operators. Yang et al. [9] observe empirically that LLM refinement saturates in accuracy. In our framework, this saturation corresponds to Case 1 convergence: the model reaches the consensus trajectory $(s_0, (I-B)^{-1}As_0, \dots)$ and further iteration produces no change. Their observed "anchoring bias" failure mode—where the model fixates on its initial answer—corresponds to the boundary between Case 2 and Case 3, where B has an eigenvalue at 1.
Error propagation in prompt composition. Kim et al. [7] bound error propagation through composed transformer layers via their Composition Lemma: $\|G_{1:T} - \tilde{G}_{1:T}\| \leq \sum_t \varepsilon_t \prod_{s>t} L_s$. This is a special case of our Lipschitz analysis in Theorem 3.4(a), applied to a heterogeneous sequence of operators with individual Lipschitz constants L_s.

4.3. **Open questions.** We identify several directions for future work.

Conjecture 4.1 (Nonlinear extension)**.** *The convergence classification of Theorem 3.8 extends to nonlinear CoT operators $T\colon \mathcal{M}_n \to \mathcal{M}_n$ that are Lipschitz continuous. Specifically, if T is a contraction on $(\mathcal{M}_n, d)$ with Lipschitz constant strictly less than 1, then T^k converges to a unique fixed point by the Banach fixed-point theorem. The challenge lies in characterizing the analogues of Cases 2 and 3 for nonlinear operators.*

(1) **Stochastic operators.** LLM outputs are stochastic. Extending the framework to random CoT operators, where each application of a prompt produces a random transformation, would require the theory of random products of matrices [3]. The relevant quantity becomes the Lyapunov exponent $\lambda = \lim_{k\to\infty} \frac{1}{k} \log \|B_k \cdots B_1\|$, and convergence holds when $\lambda < 0$.
(2) **Joint spectral radius computation.** For heterogeneous prompt sequences (Proposition 3.12), convergence is controlled by the joint spectral radius $\hat{\rho}(\{B_p\})$. Computing $\hat{\rho}$ is NP-hard in general. Developing practical approximation algorithms tailored to the CoT setting is an important open problem.
(3) **Optimal prompt ordering.** Given a set of prompts $\{p_1, \dots, p_k\}$, what ordering minimizes the number of iterations to convergence? This is related to optimal control of semigroup dynamics and appears computationally difficult in general.

(4) **Approximate commutativity.** When operators "nearly commute" ($\|[T_1, T_2]\| < \varepsilon$), how does this affect convergence? Quantitative perturbation bounds relating the commutator norm to the deviation of convergence rates from the commutative case would be valuable.
(5) **Infinite-dimensional extension.** Extending the analysis from $E = \mathbb{R}^d$ to infinite-dimensional spaces (e.g., L^2 spaces of token distributions) would require careful treatment of compactness conditions, following the approach of Gerlach and Gluck [4] for AM-compact operators.

4.4. **Limitations.** Our framework rests on a linear model of CoT operators ($s_i \mapsto As_{i-1} + Bs_i$), whereas real LLM transformations are highly nonlinear. The linear theory applies rigorously to the linearized dynamics near fixed points: if a nonlinear CoT operator has a fixed trajectory τ^*, then the Jacobian at τ^* has precisely the block lower-triangular structure of $M(A, B)$, and our spectral criteria govern local convergence. For global convergence of nonlinear operators, contractivity must be established by other means (e.g., Lyapunov functions as in Akyildiz et al. [1]).

The deterministic framework does not account for the stochastic nature of LLM outputs. While our results describe the "expected" behavior of a deterministic linearization, real systems exhibit variance that can affect convergence. The framework also assumes a fixed trajectory length n; extending to variable-length trajectories requires additional structure on the metric space.

Finally, our computational verification uses finite-dimensional examples with $d \leq 3$. Real embedding spaces have $d \sim 10^3$–10^4, and the spectral properties of the effective B-matrix in these high-dimensional settings remain to be investigated empirically.

5. Conclusion

We have established the first rigorous algebraic framework for chain-of-thought prompt composition, modeling CoT operators as elements of a semigroup acting on a metric space of reasoning trajectories. The five main results of this paper—the monoid structure (Theorem 3.1), the sharp contractivity criterion (Theorem 3.4), the idempotent–fixed point correspondence (Theorem 3.6), the four-case convergence classification (Theorem 3.8), and the JdLG decomposition (Theorem 3.10)—provide a complete algebraic characterization of when and how iterative CoT prompting converges.

The central insight is that convergence of iterated CoT application is governed by a single computable quantity: the spectral radius $\rho(B)$ of the current-state transformation matrix. When $\rho(B) < 1$, iterates converge exponentially to a consensus trajectory determined by the initial query; when $\rho(B) = 1$, the system oscillates or stabilizes depending on the spectral structure of B; and when $\rho(B) > 1$, the system diverges. The JdLG decomposition further separates the trajectory space into stable reasoning patterns and transient noise, providing a principled notion of "signal versus noise" in reasoning trajectories.

This framework opens several directions for future investigation, including extension to nonlinear and stochastic operators, computation of joint spectral radii for heterogeneous prompt sequences, and empirical validation in high-dimensional embedding spaces. We anticipate that the algebraic perspective developed here will provide useful tools for the principled design and analysis of iterative reasoning systems.

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